FF-H-116E
March 21, 1977
SUPERSEDING
Int. Fed. Spec. FF-H-00116D(CSA-FSS)
May 5, 1971 and
Fed. Spec. FF-H-116C
July 3, 1957

FEDERAL SPECIFICATION

HINGES, HARDWARE, BUILDERS'

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. The hardware covered by this specification is generally classified as hinges and spring hinges. Hinges for marine use and furniture are not included.

1.2 Classification. Hinges furnished under this specification shall be of the types and sizes specified (see 6.2) as described in section 3.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

PPP-H-1581 - Hardware (Fasteners and Related Items) Packaging and Packing for Shipment and Storage of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American National Standards Institute (ANSI) Standard:

A156.1 - Butts and Hinges.
A156.7 - Template Hinge Dimensions.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Applicable requirements. Hinges corresponding to types covered by ANSI A156.1 shall meet all requirements of A156.1. Hinges other than types covered by A156.1 shall meet requirements as herein stated. All hinges supplied shall meet the general requirements of this specification and any additional requirements stated in the procurement documents.

3.2 Materials, general.

3.2.1 Optional materials. Where materials specified herein are designated as given below, the respective optional materials listed may be substituted. (This option does not apply to finish (see 3.4.1).)

Designated material:	Optional material:
Brass	Bronze. (See 3.3.1.)
Bronze	Brass. (See 3.3.1.)
Wrought (metal)	Rolled, forged, extruded, or pressure cast forms of the metal specified.

3.2.2 Stainless steel. Stainless steel hinges shall be 300 series. Stainless steel pins shall be a commercial "18-8" corrosion-resisting steel alloy containing not less than 16 percent chromium and not less than 7 percent nickel (see 3.10.3).

3.3 Brass and bronze.

3.3.1 Chemical composition. Brass and bronze shall be suitable alloys composed chiefly of copper, with other elements as needed to provide desired appearance, and the strength, hardness and other desirable properties usually associated with red bronze and yellow brass.

3.4 Finishes, general.

3.4.1 Standard finishes. Exposed surfaces of hardware shall have the applicable standard finishes specified in the bidding documents (see 6.2, 6.5.3, and table III). If no finish is specified in the bidding documents, hinges shall be finished in accordance with 3.4.2.

3.4.2 Regular finishes. Iron and steel shall be plated and have US 10 finish; brass shall have US 4 finish; bronze shall have US 10 finish; aluminum shall have US 28 finish; and stainless steel shall have US 32D finish.

3.4.3 Plated, natural, and statuary finish hardware shall have a coating of transparent lacquer except that bright chromium, bright nickel, and zinc plated or zinc coated finishes may be furnished without lacquer.

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3.4.4 Finishes on steel. Plated steel with plated finishes shall be polished or highly polished as specified herein for the respective types. Bright surfaces which are highly polished shall have highly reflective surfaces showing maximum smoothness, evenness, and uniformity such as produced by grinding, polishing, and buffing; dull finishes which are highly polished shall be produced similarly and be uniform in texture. Polished finishes shall be applied to surfaces that are suitably prepared by rolling or forging, with or without subsequent grinding or polishing so as to produce a bright or dull finish as required. For purposes of this specification, significant surfaces shall be as defined in ANSI A156.1.

3.5 Chromium plating, finishes US26 and US26D.

3.5.1 Chromium plating shall be applied over a coating of nickel as specified under 3.5.2 and 3.5.3. On significant surfaces, the minimum thickness of the chromium shall be at least one-hundred thousandth inch (0.00001 inch).

3.5.2 Nickel plating on brass and bronze. The nickel plating shall be applied directly to brass and bronze. On significant surfaces, the minimum thickness of the nickel coating shall be at least one ten-thousandths inch (0.0001 inch).

3.5.3 Nickel plating on steel. The nickel plating shall be applied either directly or over a coating of copper. On significant surfaces, the minimum thickness of the nickel coating or of the combined layers of copper and nickel shall be at least four ten-thousandths inch (0.0004 inch), and when copper is used, the minimum thickness of the final nickel layer shall be at least two ten-thousandths inch (0.0002 inch).

3.6 Zinc coating, finishes US2G and US2H.

3.6.1 On significant surfaces, the minimum thickness of the zinc coating shall be at least five ten-thousandths inch (0.0005 inch) (see 4.2 and 4.4).

3.6.2 Application of zinc coating. Zinc coating (galvanizing) shall be applied either by hot-dipping in molten zinc or by electroplating, unless the hot-dip coating, finish US2H, is specified. However hot-dipping shall not be employed for coating alloy or heat-treated steel that may be injuriously affected at the temperature of the molten zinc; for such steel, the coating shall be applied by electroplating (see 6.2).

3.7 Primed for painting,--finish USP. Hinges primed for painting shall have an adherent paint coating of neutral color that forms a suitable undercoat for subsequent painting. This coating shall be applied to steel or iron which has been suitably cleaned and phosphatized.

3.8 Marking. All hinges other than spring, strap, and tee hinges shall bear the name or trademark of the manufacturer.

3.9 Swaging. The clearance between the leaves of full mortise hinges when closed to the parallel position shall be 1/16 inch.

3.10 Hinge pins.

3.10.1 Lubricating of pins. All hinge pins listed in ANSI A156.1 shall be effectively lubricated with a non-corrosive, nonfluid grease.

3.10.2 Nonrising loose pins. Shall be retained by a means that offers effective resistance to movement of the pin from its seat, but allows the pin to be removed by a moderate force. Pins shall be removable without damaging exposed surfaces of the hinge or tip.

3.10.2.1 Non-removable pins. Hinges for locked doors opening out shall have non-removable pins (NRP).

3.10.3 Material of pins. All non-ferrous hinges shall be supplied with corrosion-resisting steel pins as a standard. Protruding heads of corrosion-resisting steel pins shall be finished to match the hinge.

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3.10.4 Button tips. Button tips shall be of the cylindrical type, having a square shoulder approximately 1/8 inch high which aligns flush with the barrel of the hinge. End surfaces of the buttons shall be flat or convex in shape.

3.10.5 Hospital tips. Hospital tips shall be furnished when the type number specified so indicates.

3.11 Screw holes. All screw holes shall be countersunk. Screw holes shall conform to all applicable requirements of ANSI A156.7. The countersunk portion of screw holes in hinges shall form a good seat for the screw heads, with no sharp edges at the back.

3.12 Fasteners. Hinges furnished under this specification shall be complete with all screws necessary for proper installation. Such fasteners shall be of suitable size and type and shall harmonize with the material and finish of the hinge.

3.13 Workmanship. All hardware shall be free from imperfections and defects which affect its appearance, durability, and serviceability. All finished parts of hardware that are to be set flush shall be accurately machined, true edges. Edges of unfinished steel and unpolished work shall not be machine finished, but shall be smooth and reasonably true.

3.14 Hinge types.

3.14.1 Hinges, general. Hinge types listed in table I shall meet the requirements for their corresponding designations as detailed in ANSI Standard A156.1.

3.14.1.1 Sizes. Hinge sizes are available as detailed in ANSI A156.1 and should be fully specified (see 6.2 and 6.6.1).

TABLE I. General hinges

Classification	Corresponding ANSI A156.1 designation	
Types		
2102	A2112	Polished, plated.
T2102		Polished, plated, templated.
TH2102		Polished, plated, templated, hospital tip.
2105	A8112	Polished, plated.
T2105		Polished, plated, templated.
TH2105		Polished, plated, templated, hosptial tip.
2106	A2112	Phosphatized, primed.
T2106		Phosphatized, primed, templated.
2107	A8112	Polished, plated.
T2107		Polished, plated, templated.
TH2107		Polished, plated, templated, hospital tip.
2112	A2111	Highly polished, plated.
T2112		Highly polished, plated, templated.
TH2112		Highly polished, plated, templated, hospital tip.
2115	A8111	Highly polished, plated.
T2115		Highly polished, plated, templated.
TH2115		Highly polished, plated, templated, hospital tip.
2117		Phophatized, primed.
T2117		Phophatized, primed, templated.
TH2117		Phosphatized, primed, templated, hospital tip.
T2119	A8161	Highly polished, plated, templated.
TH2119		Highly polished, plated, templated, hospital tip.
T2121		Phosphatized, primed, templated.
TH212		Phosphatized, primed, templated, hospital tip.
2122	A2133	Polished, plated.
T2122		Polished, plated, templated.
2125	A8133	Polished and plated.
T2125		Polished and plated, templated.
2126	A2133	Polished and plated.
2127	A8133	Phosphatized and primed.
T2127		Phosphatized and primed, templated.
2127H		US2H plated.
2132	A2412	Highly polished, plated.
T2132		Highly polished, plated, templated.
TH2132		Highly polished, plated, templated, hospital tip.

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TABLE I. General hinges (con.)

Classification	Corresponding ANSI A156.1 designation	
Types		
2133	A8412	Highly polished, plated.
T2133		Highly polished, plated, templated.
TH2133		Highly polished, plated, templated, hospital tip.
2135		Phosphatized and primed.
T2135		Phosphatized and primed, templated.
TH2135		Phosphatized and primed, templated, hospital tip.
T2136	A2411	Highly polished, templated.
TH2136		Highly polished, templated, hospital tip.
T2137	A8411	Highly polished, templated.
TH2137		Highly polished, templated, hospital tip.
T2139		Phosphatized, primed, templated.
TH2139		Phosphatized, primed, templated, hospital tip.
T2142	A2212	Polished, plated, templated.
TH2142		Polished, plated, templated, hospital tip.
T2143	A8212	Polished, plated, templated.
TH2143		Polished, plated, templated, hospital tip.
T2145	A8212	Phosphatized and primed, templated.
TH2145		Phosphatized and primed, templated, hospital tip.
T2146	A2211	Highly polished, plated, templated.
TH2146		Highly polished, plated, templated, hospital tip.
T2147	A8211	Highly polished, plated, templated.
TH2147		Highly polished, plated, templated, hospital tip.
T2149		Phosphatized and primed, templated.
TH2149		Phosphatized and primed, templated, hospital tip.
T2152	A2312	Highly polished, templated.
TH2152		Highly polished, templated, hospital tip.
T2153	A8312	Highly polished, plated, templated.
TH2153		Highly polished, plated, templated, hospital tip.
T2155		Phosphatized and primed, templated.
TH2155		Phosphatized and primed, templated, hospital tip.
T2156	A2311	Highly polished, plated, templated.
TH2156		Highly polished, plated, templated, hospital tip.
T2157	A8311	Highly polished, plated, templated.
TH2157		Highly polished, plated, templated, hospital tip.
T2159		Phosphatized and primed, templated.
TH2159		Phosphatized and primed, templated, hospital tip.
T2160	A8221	Highly polished, templated.
T2161		Phosphatized and primed, templated.
T2163	A8321	Highly polished, templated.
T2164		Phosphatized and primed, templated.
2165	A8381	Phosphatized and primed, with screw holes.
2165A		Phosphatized and primed, without screw holes.
2167	A8391	Phosphatized and primed, with screw holes.
2167A		Phosphatized and primed, without screw holes.

3.14.2 Spring hinges. Spring hinges furnished under this specification shall conform to the requirements stated for individual types.

3.14.2.1 Type 2301. Full surface, wrought steel, plated, steel pin. Size 3 inches x 3 inches. Coil spring is fully enclosed and with adjustable tension. Fast pin, button tip. Thickness of metal shall be 0.082 ± 0.005 inches. Wood screw fasteners shall be furnished. Hinges shall be reversible.

3.14.2.2 Type 2302. Half surface, wrought steel, plated, steel pin. Size (length) 3 inches. Construction same as 2301. Width of flange shall be not less than 1-3/8 inches from the center of the pin to the edge of the flange.

3.14.2.3 Type 2331A. Butt, single acting, wrought steel, plated, steel pin. For application to hanging strip.

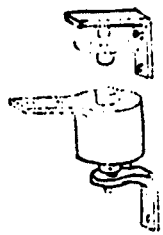
3.14.2.4 Type 2333A. Same as 2331A except for application without hanging strip.

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Sizes for 2331A and 2333A

Size Length of flange	Thickness of flange minimum	Dia. of Barrell minimum	Application of hinges - door size Thickness of door	Max width of door
Inch	Inch	Inch	Inch	Inch
5	.091	.98	1-1/8 to 1-1/2	30
6	.102	1.09	1-1/4 to 1-3/4	32
8	.128	1.37	1-1/2 to 2-1/4	34

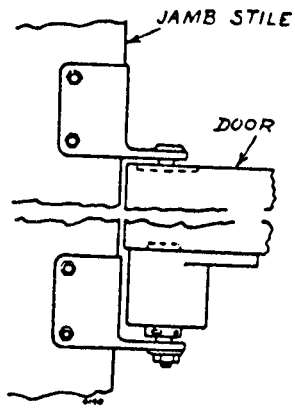
3.14.2.5 Type 2390. Pivot design, double acting. For gates or dwarf doors 1-1/8 to 2-1/2 inches thick. Exposed parts shall be of bronze. Steel coil spring with external spring tension adjustment. Ball-bearing bottom pivot. Adjustment device shall provide for alignment of door and for holding door closed or open at any desired position. When installed, this hinge projects approximately 5 inches below the bottom edge of the door.



3.14.2.6 Type 4140, lavatory hinge. For lavatory partition doors of 7/8 inch or greater finished thickness. Pivot type spring hinge for single or double-acting doors. Fully enclosed steel spring with external adjustment. Ball bearing bottom pivot. Adjustable door alignment. Spring case 2-3/8 inches to 2-5/8 inches in diameter. Clamp flanges for jamb stile 1-1/4 inches thick, for mounting above and below the door, and not to extend more than 4 inches beyond the door. Hinge to have adjustment for holding door at any point of swing; shall be set to hold door open unless otherwise specified.

Fasteners for flanges: Two screws in each flange shall be either 1/4 - 20 brass threaded rod with hexagonal rounded closed cap nut on each end or 1/4-20 through bolts with hexagonal rounded head and closed cap nut.

Fastenings for doors: Flat plates applied to door with flat-head wood or machine screws to suit application.

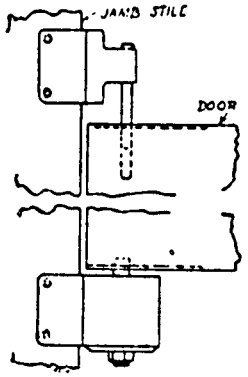


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3.14.2.7 Type 4200, lavatory hinge. For lavatory partition doors other than metal of 7/8 inch or greater finished thickness. Pivot type gravity hinge for single or double-acting doors. Ball bearing bottom pivot. Clamp flanges for jamb stile 1-1/4 inches thick, for mounting above and below door and to extend not more than 4 inches beyond door. Hinge shall be fully encased and provide adjustment for holding the door at any point of swing, or against stop, with smooth, even motion. Exposed parts except upper pivot pin shall be of chromium plated brass. Upper pivot pin shall be of stainless steel and shall operate in a bronze bearing. Ball bearings, raceway, and cam shall be of hardened stainless steel.

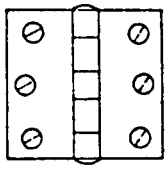
Fasteners for flanges: Two screws in each flange, either 1/4-20 brass threaded rod with hexagonal rounded closed cap nut at each end; or 1/4-20 through bolts with hexagonal rounded head and closed cap nuts for ends.

Fastening for door: Flat plates applied to door with flat head wood screws.



3.14.3 Miscellaneous hinges. Miscellaneous hinges furnished under this specification shall conform to the following requirements:

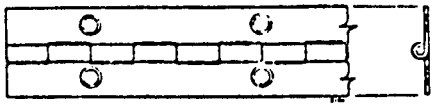
3.14.3.1 Type 2171H. Full mortise, regular weight, five knuckles, fast pin, unpolished wrought steel, zinc coated, finish US2G, brass pin.



Sizes

Size, open	Thickness of metal ± 0.005 inch
Inches	Inch
2 x 2	0.083
2-1/2 x 2-1/2	.089
3 x 3	.092
3-1/2 x 3-1/2	.123
4 x 4	.130

3.14.3.2 Type 2174F. Continuous wrought steel with steel pin. Steel pin 0.090 inch diameter. Screw holes on 2 inch centers, countersunk use No. 4 screws.



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Sizes			
Width open	Thickness of Metal ± 0.005	Length of joint	Length
Inch	Inch	Inch	Inches
1-1/16	0.045 ± 0.005	5/8	24, 30, 36, 48, 60, 72
1-1/4	.045 ± .005	5/8	24, 30, 36, 48, 60, 72
1-1/2	.045 ± .005	5/8	24, 30, 36, 48, 60, 72

When specified, furnish without screw holes and screws.

3.14.3.3 Type 2175. Continuous wrought brass, brass pin 0.128 inch diameter. Screw holes countersunk on 2 inch centers use no. 6 screws. Design same as type 2174F.

Sizes			
Width open	Thickness of metal ± 0.005	Length of joint	Length
Inch	Inch	Inch	Inches
1-1/4	0.062	3/4 ± 0.005	72
1-1/2	.062	3/4 ± 0.005	72
2	.062	3/4 ± 0.005	72

When specified, furnish without screw holes and screws.

3.14.3.4 Type 2201. Strap, light, wrought steel, zinc coated, US2G finish with brass or stainless steel pin.



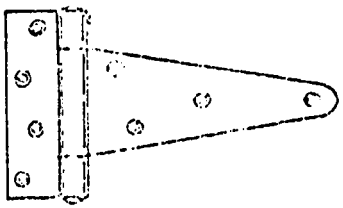
Sizes		
Length Closed	Width of strap of joint	Minimum thickness of metal
Inches	Inches	Inch
3	1-3/32	0.058
4	1-1/8	.058
5	1-1/4	.069
6	1-1/2	.083

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3.14.3.5 Type 2203. Strap, heavy, wrought steel, hot-dip zinc coated US2G finish with brass or stainless steel pin.

Sizes		
Length closed	Width of strap of joint	Minimum thickness of metal
Inches	Inches	Inch
4	1-9/16	0.080
5	1-3/4	.091
6	2-1/2	.115
8	3	.125

3.14.3.6 Type 2207. Tee, light, wrought steel, zinc coated, US2G finish, brass or stainless steel pin.

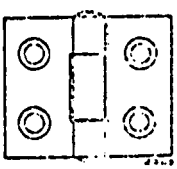


Sizes				
Length closed	Width of strap of joint	Length of joint	Width of pad	Minimum thickness of metal
Inches	Inches	Inches	Inches	Inch
3	1	2-1/4	3/4	0.062
4	1-1/16	2-1/2	3/4	.062
6	1-1/4	3	7/8	.068

3.14.3.7 Type 2208. Tee, heavy, wrought steel, zinc coated, US2G finish, brass or stainless steel pin.

Sizes				
Length closed	Width of strap of joint	Length of joint	Width of pad	Minimum thickness of metal
Inches	Inches	Inches	Inches	Inch
4	1-1/8	2-1/8	7/8	0.068
6	1-1/2	3-1/2	1-3/16	.088
8	1-5/8	3-7/8	1-1/4	.092

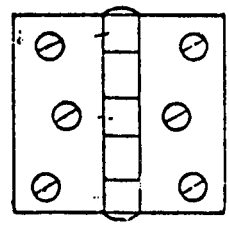
3.14.3.8 Type 2269. Narrow, fast-pin, for cabinets, full mortise, three or five knuckles, wrought steel, US2G finish, brass or stainless steel pin with spun tips.



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Sizes		
Length of joint	Width open	Thickness of metal
Inches	Inches	Inch
1	1-3/16	0.056
1-1/2	1-1/2	.065

3.14.3.9 Type 2171H. Transom hinge. Wrought steel, zinc coated, US2G finish, brass pin, five knuckles.



Nontemplate.
Fast pin.

Sizes	
Size, open	Thickness of metal ± 0.005 inch
Inches	Inch
2 x 2----------	0.083
2-1/2 x 2-1/2----------	.089
3 x 3----------	.092
3-1/2 x 3-1/2----------	.123
4 x 4----------	.130

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Visual and dimensional examination. Hinges furnished under this specification shall be examined for the defects listed in table II. The inspection level shall be level II of MIL-STD-105, with an acceptable quality level (AQL) of 6.5 for total defects; expressed in terms of defects per hundred units.

TABLE II. Defects

(a)	Hardware not conforming to specified sizes (see 3.1 and applicable section of 3.14).
(b)	Material not as specified (see 3.2 and 3.3).
(c)	Finish not as specified (see 3.4, 3.5, 3.6, and 3.7).
(d)	Hinges, other than strap, spring, and tee not properly marked (see 3.8).
(e)	Clearance between leaves of mortise hinges not 1/16 inch (see 3.9).
(f)	Improper lubrication of hinge pins (see 3.10.1).
(g)	Non-rising loose pin not retained to seat (see 3.10.2).
(h)	Pins of non-ferrous hinges not of corrosion-resisting steel (see 3.10.3).
(i)	Button tips not in accordance with 3.10.4.
(j)	Hospital tips not furnished when required (see 3.10.5).
(k)	Screw holes not in accordance with 3.11.
(l)	Fasteners not furnished in accordance with 3.12.
(m)	Workmanship not as specified (see 3.13).

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4.3 Finish and performance. When specified (see 6.2), samples of hinges from each lot shall be selected in accordance with inspection level S-1 of MIL-STD-105. Samples shall be tested in accordance with 4.4 and 4.5. The acceptable quality level (AQL) shall be 2.5; expressed in terms of defects per hundred units.

4.4 Finish testing. Finishes of hinges shall be tested in accordance with the procedure detailed in ANSI A156.1.

4.5 Performance testing. Hinges listed in table I of this specification shall be tested in accordance with the applicable sections of ANSI A156.1 (see 3.14.1 and table I).

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Hinges shall be preserved, packaged, packed, and marked in accordance with applicable levels of PPP-H-1581. (For levels A packaging and packing.)

5.2 Packaging.

5.2.1 Commercial. The hinges shall be packaged in unit containers normally used in commercial practice.

5.3 Packing.

5.3.1 Commercial. The hinges shall be packed in shipping containers to insure safe delivery at destination, to provide for safe redistribution by the initial receiving activity, and shall be acceptable by common carrier under National Motor Freight Classification or Uniform Freight Classification.

5.4 Unitization. (Applicable to full rail car and truck load shipments only.) The hinges shall be unitized for shipment in accordance with normal commercial practice. The unitized load shall not exceed 2,500 lbs. in weight.

5.5 Marking. Packages, shipping containers, and unitized loads (when applicable) shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Application. Hinges covered by this specification are intended for a wide range of door and lid mountings. Purchasers should consider the usage for which the hinge is intended in making selections.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size of hinge required (see 1.2 and applicable section of 3.14).
(c) If finishes other than regular finishes are required (see 3.4.1).
(d) If electroplating is required in zinc finish (see 3.6.2).

6.3 Dimensions. Where more than one dimension is given, the vertical dimension is given first.

6.4 Handing, right and left.

6.4.1 Rules for hands of hinges. The hand is determined from the closed door when facing the door from the hinge pin side. If the hinges are on the viewers right they are right-hand hinges. If they are on the left, they are left-hand hinges.

6.4.2 Rules for hand of lavatory doors.

6.4.2.1 The hand of doors is determined from the outside, which for lavatory doors is the room side. If the hinges are on the left it is a left-hand door; if on the right, it is a right-hand door.

6.4.2.2 Single-acting spring hinges are regularly made to close the door and when made to open the door are designated "reverse acting".

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6.5 Finishes.

6.5.1 Preferred finishes. As far as is practicable, finishes should be confined to prime painted, dull bronze, chromium-plated, and zinc coated.

6.5.2 Selection of finishes. Where finishes other than those given above are required, selections should be confined to standards adopted by the manufacturers as listed in table III.

6.5.3 Standard finishes. The standard finishes for builders' hardware are intended primarily to promote uniformity in the appearance of builders' hardware items, and for the standard finishes subject to undesirable variations in appearance, duplicate samples are used by the manufacturers as a guide for color and texture.

The "BHMA Equivalents" included in table V are those detailed in Builders' Hardware Manufacturers Association Standard 1301. They are included for the convenience of the manufacturer.

TABLE III. Standard finishes

Symbol	BHMA Equivalents	General Description	Metal applied to
USP	600	Primed for painting	Steel
US1B	601	Bright Japanned	Steel
US1D	None	Dead black	Steel
US2C	602	Cadmium plated*	Steel
US2G	603	Zinc plated	Steel
US2H	None	Zinc plated, hot-dipped	Steel
US3	605	Bright brass, clear coated	Brass
US4	606	Stain brass, clear coated	Brass
US10	612	Satin bronze, clear coated	Bronze
US14	618	Bright nickel plated, clear coated	Brass, bronze
US26	625	Bright chromium plated	Brass, bronze
US26D	626	Satin chromium plated	Brass, bronze
US27	627	Satin aluminum, clear coated	Aluminum
US28	628	Satin aluminum, clear anodized	Aluminum
US32	629	Bright stainless steel	Stainless steel
US32D	630	Satin stainless steel	Stainless steel

*Either cadmium or zinc, electroplated, may be furnished for US2C.

6.5.4 Comparison of standard finishes. Owing to differences in the processes of producing hardware finishes and the variety of metals to which they are applied, it is commercially impracticable to attain an exact match. Therefore, it is understood that hardware delivered in a standard finish from two or more sources will compare reasonably when the items are viewed at arm's length and approximately two feet apart. This approximates the conditions of use, as for example, hinges and lock trim on opposite edges of a door.

6.5.5 Finishes for lavatory spring hinges. On brass and bronze, the finishes should be limited to chromium plated, US26, and US26D.

6.6 Lavatory clamp fittings. Clamp fittings specified herein to be nonadjustable are intended for 1-1/4 inch partitions and jamb stiles subject to variations of not more than 1/32 inch in thickness. Adjustable clamp fittings, providing for greater variations in thickness, are optional with the supplier on hinges and stops for mounting on jamb stiles. Where partitions and jamb or door stiles are of thickness other than 1-1/4 inches, it should be so specified.

6.7 Definitions of terms. There are given below definitions of full mortise, full surface, half surface and half mortise hinges as generally applied by the trade and as used in this specification.

Full Mortise. A term applied mainly to hinges which are arranged for attachment entirely by mortising the door leaf and the jamb leaf (see 1.2).

Full Surface. A term applied mainly to hinges which are arranged for attachment entirely on the surface of the door and jamb without mortising (see 1.2).

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Half Surface. A term applied mainly to hinges designed for surface application of the door leaf and for mortising the jamb leaf (see 1.2).

Half Mortise. A term applied mainly to hinges designed for mortise application of the door and surface to the jamb (see 1.2).

6.8 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Government requirements.

6.9 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.10 Information on template hinges is contained in ANSI A156.7.

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